Introduction to Mathematics for Deep Learning: Homeworks #3 Solutions

1. Let $X \in \mathbb{R}^n$ be a random vector with mean $\boldsymbol{\mu} = \mathbb{E}[X]$. Recall that the covariance matrix is defined by

$$\mathrm{Cov}(X) \;=\; \mathbb{E}\big[(X-\boldsymbol{\mu})(X-\boldsymbol{\mu})^\top\big]\,.$$

Show that

$$\mathrm{Cov}(X) \;=\; \mathbb{E}\big[XX^\top\big] \;-\; \boldsymbol{\mu}\boldsymbol{\mu}^\top.$$

(Compare with the scalar identity $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.)

Solution:
Starting from the definition and expanding:

$$\begin{aligned}
\mathrm{Cov}(X) &= \mathbb{E}\big[(X-\boldsymbol{\mu})(X-\boldsymbol{\mu})^\top\big] \\
&= \mathbb{E}\big[XX^\top - X\boldsymbol{\mu}^\top - \boldsymbol{\mu}X^\top + \boldsymbol{\mu}\boldsymbol{\mu}^\top\big] \\
&= \mathbb{E}[XX^\top] - \mathbb{E}[X]\boldsymbol{\mu}^\top - \boldsymbol{\mu}\,\mathbb{E}[X]^\top + \boldsymbol{\mu}\boldsymbol{\mu}^\top \\
&= \mathbb{E}[XX^\top] - \boldsymbol{\mu}\boldsymbol{\mu}^\top,
\end{aligned}$$

where we used linearity of expectation and $\mathbb{E}[X] = \boldsymbol{\mu}$. This proves the identity.

2. (a) For positive semidefinite matrix A, show that all its eigen values are nonnegative
   (b) Show that a matrix A is positive semidefinite if and only if there exists a matrix B such that $A = B^T B$.
   (c) For a matrix A and positive semidefinite matrix Φ, show that $\nabla_\Phi \mathrm{tr}(A\Phi) = A^T$.

Solution:

(a) Suppose there exists a negative eigenvalue λ. Let v be the corresponding eigenvector, then

$$v^T A v = \lambda v^T v < 0 \tag{1}$$

which is a contradiction to the positive semidefinite assumption.

(b) If $A = B^T B$, then for all v, we have

$$v^T A v = \|Bv\|^2 \geq 0. \tag{2}$$

On the other hand, let $A = Q\Lambda Q^T$ be the eigen decomposition of matrix A. Since all its eigen values are nonnegative, we can let

$$B = Q\sqrt{\Lambda} \tag{3}$$

where $\sqrt{\Lambda}$ is a diagonal matrix with all its entries are square root of eigenvalues $\sqrt{\lambda_i}$'s.

(c) Let $C = A\Phi$, then

$$\mathrm{tr}(C) = \sum_{i=1}^{d}\sum_{j=1}^{d} A_{i,j}\Phi_{j,i}. \tag{4}$$

Thus,

$$\frac{\partial}{\partial \Phi_{i,j}}\mathrm{tr}(C) = A_{j,i}. \tag{5}$$

Thus, $\nabla_{\Phi}\mathrm{tr}(A\Phi) = A^T$.

3. Let $X \in \mathbb{R}^{n\times n}$ be *positive definite* (so $\log|X|$ is well-defined).

(a) Let adjugate of matrix A be $\mathrm{adj}(X)$. Show that

$$\nabla_X |X| = \mathrm{adj}(X)^{\intercal}$$

(b) Compute the gradient of $\log|X|$ with respect to X.

Solution:
We use the cofactor (adjugate) expansion of the determinant.

(a) Fix indices i, j. Let $E^{(ij)}$ be the matrix with a 1 at position (i, j) and zeros elsewhere, and define

$$g_{ij}(t) \;:=\; \det\big(X + t\,E^{(ij)}\big).$$

By cofactor expansion along the j-th column (or i-th row), the determinant is an affine function of the entry X_{ij} with slope equal to the cofactor $C_{ij}(X)$:

$$g_{ij}(t) \;=\; \det(X) \;+\; t\,C_{ij}(X) \quad\Rightarrow\quad \frac{\partial}{\partial t}\det\big(X + t\,E^{(ij)}\big)\bigg|_{t=0} \;=\; C_{ij}(X).$$

Therefore,

$$\frac{\partial\,|X|}{\partial X_{ij}} \;=\; C_{ij}(X).$$

Collecting all partials, the gradient of the determinant is the transpose of the cofactor matrix:

$$\nabla_X |X| \;=\; \big[C_{ij}(X)\big]_{ij}^{\top} \;=\; \mathrm{adj}(X)^{\top}.$$

(b) By the scalar chain rule,

$$\frac{\partial}{\partial X_{ij}}\log|X| \;=\; \frac{1}{|X|}\,\frac{\partial|X|}{\partial X_{ij}} \;=\; \frac{1}{|X|}\,C_{ij}(X).$$

Hence, in matrix form,

$$\nabla_X \log |X| \;=\; \frac{1}{|X|} \operatorname{adj}(X)^\top.$$

For any invertible X, $\operatorname{adj}(X) = |X|\, X^{-1}$. Therefore,

$$\nabla_X \log |X| \;=\; \frac{1}{|X|} \left(|X|\, X^{-1}\right)^\top \;=\; X^{-T}.$$

Since X is symmetric positive definite, $X^{-T} = X^{-1}$, and we conclude

$$\boxed{\nabla_X \log |X| \;=\; X^{-1}}.$$

4. For $A \in \mathbb{R}^{n \times m}$, show that the orthogonal complement of row space is a null space of A.

Solution:
Let

$$A = \begin{bmatrix} a_1^T \\ a_2^T \\ a_n^T \end{bmatrix}. \tag{6}$$

Then, the null space of A is a set of x such that

$$Ax = \begin{bmatrix} a_1^T x \\ a_2^T x \\ a_n^T x \end{bmatrix} = 0, \tag{7}$$

i.e.,

$$null(A) = \{x : a_i^T x \text{ for all } i\}. \tag{8}$$

On the other hand, the orthogonal complement of row space is a set of x such that

$$\left(\sum_{i=1}^{n} \alpha_i a_i^T \right) x = 0. \tag{9}$$

for all $\alpha_i \in \mathbb{R}$. This is identical to the null space of A. Formally, you have to show that

- for any $x \in null(A)$, we have $x \in rowspace(A)^\perp$
- for any $x \in rowspace(A)^\perp$, we have $x \in null(A)$

5. SKIP

6. Suppose $X|Y=0 \sim \mathcal{N}(\boldsymbol{\mu}_0, \Sigma_0)$ and $X|Y=1 \sim \mathcal{N}(\boldsymbol{\mu}_1, \Sigma_1)$, with class priors $\pi_0, \pi_1 > 0$, $\pi_0 + \pi_1 = 1$.

   (a) Show that the MAP decision rule

   $$\frac{p_{X|Y}(x|1)\pi_1}{p_{X|Y}(x|0)\pi_0} > 1$$

   leads to a quadratic decision boundary in x when $\Sigma_0 \neq \Sigma_1$.

   (b) Show that if $\Sigma_0 = \Sigma_1 = \Sigma$, the boundary becomes linear, and give its explicit equation.

   (c) Interpret the result of (b) as a special case of a Gaussian channel model, where the observation T is a Gaussian noise–corrupted version of an input x (that takes the value $\boldsymbol{\mu}_0$ or $\boldsymbol{\mu}_1$ depending on the class). Explain why, in this view, the linear decision boundary corresponds to the optimal MAP estimator for distinguishing the two possible inputs of the Gaussian channel.

Solution:
Define the log-posterior odds

$$g(x) \;:=\; \log p(x \mid 1) + \log \pi_1 - \log p(x \mid 0) - \log \pi_0.$$

Using the Gaussian density,

$$\begin{aligned} g(x) &= -\tfrac{1}{2}(x-\mu_1)^\top \Sigma_1^{-1}(x-\mu_1) - \tfrac{1}{2}\log|\Sigma_1| + \log \pi_1 \\ &\quad + \tfrac{1}{2}(x-\mu_0)^\top \Sigma_0^{-1}(x-\mu_0) + \tfrac{1}{2}\log|\Sigma_0| - \log \pi_0 \\ &= \tfrac{1}{2}x^\top(\Sigma_0^{-1} - \Sigma_1^{-1})x + x^\top(\Sigma_1^{-1}\mu_1 - \Sigma_0^{-1}\mu_0) + c, \end{aligned}$$

where

$$c = -\tfrac{1}{2}(\mu_1^\top \Sigma_1^{-1}\mu_1 - \mu_0^\top \Sigma_0^{-1}\mu_0) - \tfrac{1}{2}\log\frac{|\Sigma_1|}{|\Sigma_0|} + \log\frac{\pi_1}{\pi_0}.$$

The MAP rule predicts class 1 iff $g(x) > 0$.

   (a) If $\Sigma_0 \neq \Sigma_1$, then the quadratic term $\frac{1}{2}x^\top(\Sigma_0^{-1} - \Sigma_1^{-1})x$ is nonzero, so the decision surface $\{x : g(x) = 0\}$ is a (general) quadratic.

   (b) If $\Sigma_0 = \Sigma_1 = \Sigma$, the quadratic term cancels and

   $$g(x) = x^\top \Sigma^{-1}(\mu_1 - \mu_0) - \tfrac{1}{2}(\mu_1^\top \Sigma^{-1}\mu_1 - \mu_0^\top \Sigma^{-1}\mu_0) + \log\frac{\pi_1}{\pi_0}.$$

   Thus the boundary is affine:

   $$w^\top x \;\gtrless_{y=0}^{y=1}\; \tau, \quad w := \Sigma^{-1}(\mu_1 - \mu_0), \quad \tau := \tfrac{1}{2}(\mu_1^\top \Sigma^{-1}\mu_1 - \mu_0^\top \Sigma^{-1}\mu_0) - \log\frac{\pi_1}{\pi_0}.$$

(c) In the case $\Sigma_0 = \Sigma_1 = \Sigma$, the class label determines which mean (μ_0 or μ_1) is transmitted, and the observed vector X is corrupted by Gaussian noise:

$$T = x + N, \qquad x \in \{\mu_0, \mu_1\}, \quad N \sim \mathcal{N}(0, \Sigma).$$

This is exactly a binary-input Gaussian channel. The receiver observes T and must decide which input mean was sent. The log-likelihood ratio for this channel is

$$\log \frac{p(T \mid x = \mu_1)}{p(T \mid x = \mu_0)} = (\mu_1 - \mu_0)^\top \Sigma^{-1} T - \tfrac{1}{2}(\mu_1^\top \Sigma^{-1} \mu_1 - \mu_0^\top \Sigma^{-1} \mu_0),$$

which is an *affine function* of T—precisely the linear discriminant derived in (b). Hence the linear classifier is equivalent to the MAP detector for distinguishing two possible inputs of a Gaussian channel with equal noise covariance. This interpretation shows that linear classifier performs optimal decoding under Gaussian noise.

7. Let $X^{(1)}, \ldots, X^{(N)} \in \mathbb{R}^d$ be i.i.d. samples from $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$ with unknown mean $\boldsymbol{\mu} \in \mathbb{R}^d$ and unknown positive definite covariance $\Sigma \in \mathbb{R}^{d \times d}$. Based on MLE $\widehat{\boldsymbol{\mu}}_{\mathrm{ML}}$ and $\widehat{\Sigma}_{\mathrm{ML}}$, compute $\mathbb{E}\big[\widehat{\Sigma}_{\mathrm{ML}}\big]$ and show that the MLE covariance is biased. Propose an unbiased covariance estimator by replacing N with $N - 1$, and verify its unbiasedness.

Solution:
We have seen that

$$\widehat{\boldsymbol{\mu}}_{\mathrm{ML}} \;=\; \overline{X} := \frac{1}{N} \sum_{i=1}^{N} X^{(i)}, \qquad \widehat{\Sigma}_{\mathrm{ML}} \;=\; \frac{1}{N} \sum_{i=1}^{N} \big(X^{(i)} - \overline{X}\big)\big(X^{(i)} - \overline{X}\big)^\top.$$

Let $S := \sum_{i=1}^{N} (X^{(i)} - \overline{X})(X^{(i)} - \overline{X})^\top$. A standard identity (proved below) gives

$$\mathbb{E}[S] \;=\; (N-1)\,\Sigma \quad \Longrightarrow \quad \mathbb{E}\big[\widehat{\Sigma}_{\mathrm{ML}}\big] = \mathbb{E}\left[\frac{1}{N} S\right] = \frac{N-1}{N}\,\Sigma.$$

Hence $\widehat{\Sigma}_{\mathrm{ML}}$ is *biased low* by the factor $(N-1)/N$.

Write the centered sum decomposition

$$\sum_{i=1}^{N} (X^{(i)} - \boldsymbol{\mu})(X^{(i)} - \boldsymbol{\mu})^\top = S \;+\; N(\overline{X} - \boldsymbol{\mu})(\overline{X} - \boldsymbol{\mu})^\top.$$

Taking expectations, the left side equals $N\Sigma$; the last term has mean $\mathbb{E}\big[N(\overline{X} - \boldsymbol{\mu})(\overline{X} - \boldsymbol{\mu})^\top\big] = N \cdot \mathrm{Cov}(\overline{X}) = N \cdot (\Sigma/N) = \Sigma$. Therefore $N\Sigma = \mathbb{E}[S] + \Sigma$, i.e. $\mathbb{E}[S] = (N-1)\Sigma$.

Define

$$\widehat{\Sigma}_{\mathrm{unb}} := \frac{1}{N-1} \sum_{i=1}^{N} \big(X^{(i)} - \overline{X}\big)\big(X^{(i)} - \overline{X}\big)^\top.$$

Then

$$\mathbb{E}\big[\widehat{\Sigma}_{\text{unb}}\big] = \frac{1}{N-1}\,\mathbb{E}[S] = \frac{1}{N-1}\,(N-1)\Sigma = \Sigma,$$

so $\widehat{\Sigma}_{\text{unb}}$ is unbiased. In short, replacing N by $N-1$ debiases the MLE.